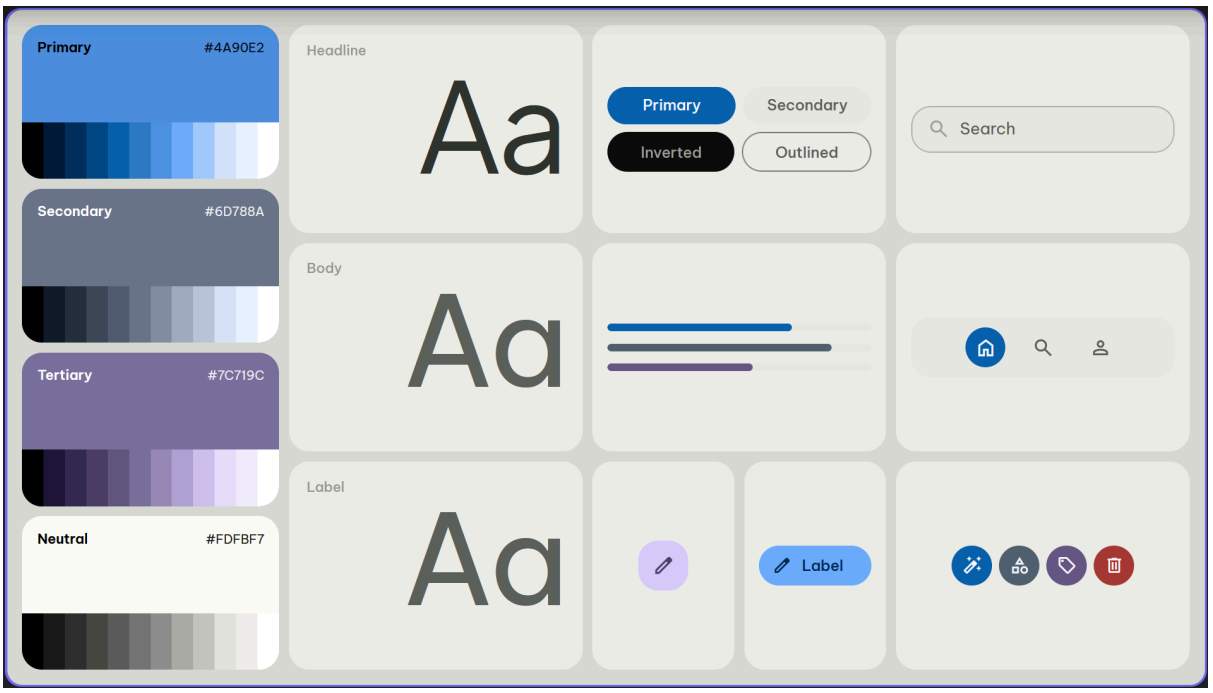
Project Scope: Baby Milestone Tracker (MVP)

1. Branding & Identity

- **App Name:** Sprout: Baby Milestone Tracker
- **Slogan:** Evidence-based peace of mind for your baby's journey.
- **Design Language:** "Clinical Calm" — Utilizing soft medical blues, warm creams, and clear, data-forward typography to evoke a sense of trust and focus.
- **Typography:** Highly legible, accessible rounded sans-serif (e.g., Nunito or SF Pro Rounded).



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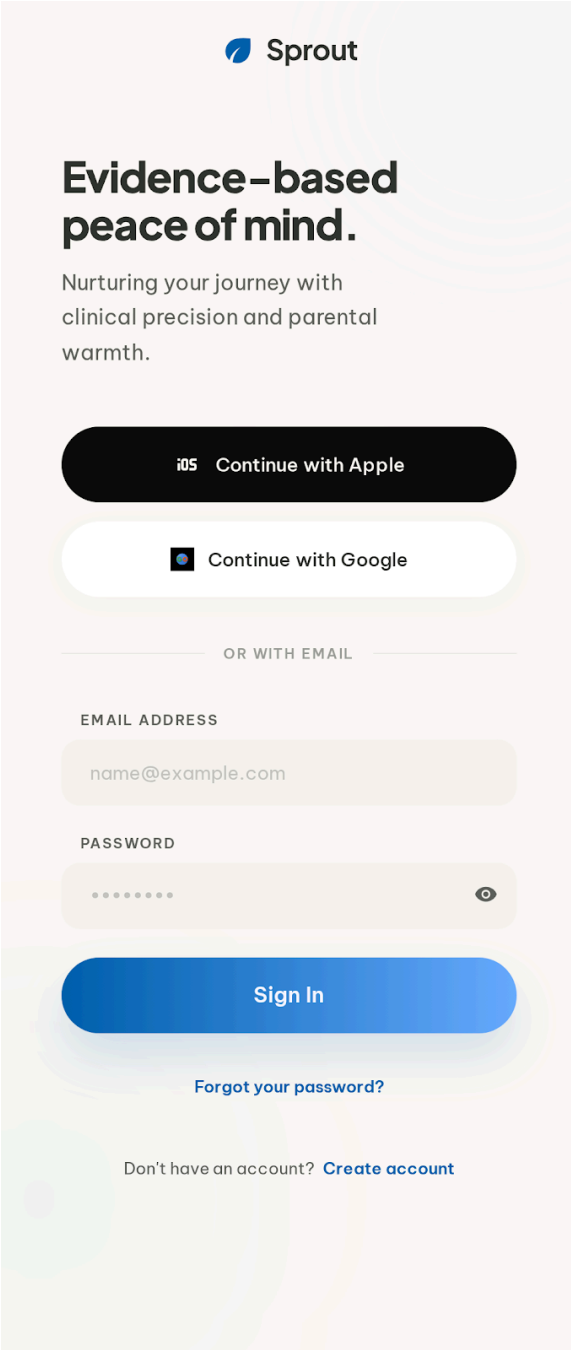
2. Functional Screen Specifications & Developer Logic

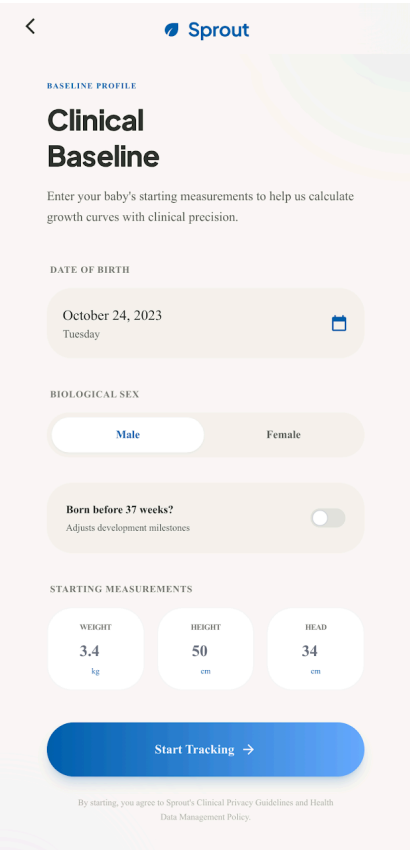
Screen 1: Login / Sign Up

- **Purpose:** Secure user onboarding and account creation.
- **Developer Notes:** Standard authentication flow.
- **Features:**
  - **Social Auth:** "Continue with Apple" (Primary) and "Continue with Google" (Secondary) to minimize friction.
  - **Manual Auth:** Standard Email and Password fields.
  - **Database Hook:** Create the initial user record in the database, generating a unique User ID.



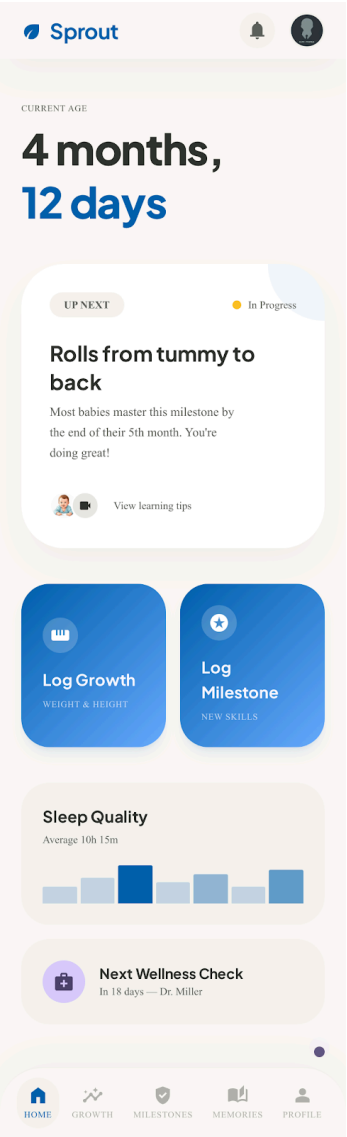
Screen 2: Baby Setup (Clinical Baseline & Initial Log)

- **Purpose:** Gather the exact parameters required to calculate WHO/CDC growth percentiles and capture the baby's current baseline to instantly populate the app's charts.
- **Developer Notes:** Upon submission, the backend must execute a batched operation: 1) Create the `Baby_Profile` record, and 2) Immediately write the first record into the `Growth_Logs` table using today's date and the provided measurements. This guarantees zero "empty states" when the user hits the main app. The algorithms strictly require biological sex and precise age to function.
- **Features:**
    - **Input - Name:** Text string.
    - **Input - Date of Birth:** Date picker. *(Crucial Dev Requirement: Calculate the baby's exact age dynamically in days, weeks, and months every time the app opens).*
    - **Input - Biological Sex:** Toggle for Male / Female. (WHO/CDC use distinct data tables based on biological sex).
    - **Input - Prematurity (Optional):** Toggle asking "Born before 37 weeks?". If yes, request the original due date to adjust chronological age for accurate tracking.
    - **Input - Current Measurements (The Baseline):** Simple input fields for Weight, Height, and Head Circumference.
    - **Action Button:** "Start Tracking" — triggers the dual database write and routes to Home.



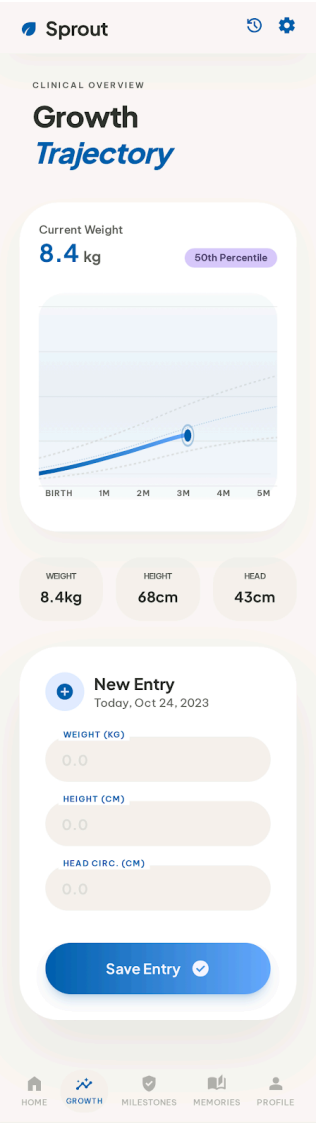
Screen 3: Home (The Reassurance Dashboard)

- **Purpose:** Provide a daily focus, immediate reassurance, and quick access to logging tools.
- **Developer Notes - Defining "Milestones":** Milestones are specific physical (motor), cognitive, social, or language skills (e.g., "Rolls over," "First smile") that act as standard checkpoints for typical development within specific age brackets.
- **Features:**
    - **Current Age Display:** Calculates and displays exact age (e.g., "4 months, 12 days").
    - **"Up Next" Logic Card:** Queries the milestone database for 1-2 skills where the `target_age_min` matches the baby's current age bracket, displaying them as upcoming goals to watch for.
    - **Visual Status UI:** On the milestone cards, display a grey outline for "Not Yet," a yellow dot for "In Progress," and a green checkmark for "Achieved."
    - **Quick Actions:** Prominent buttons to route the user to "Log Growth" (Screen 4) or "Log Milestone" (Screen 5).



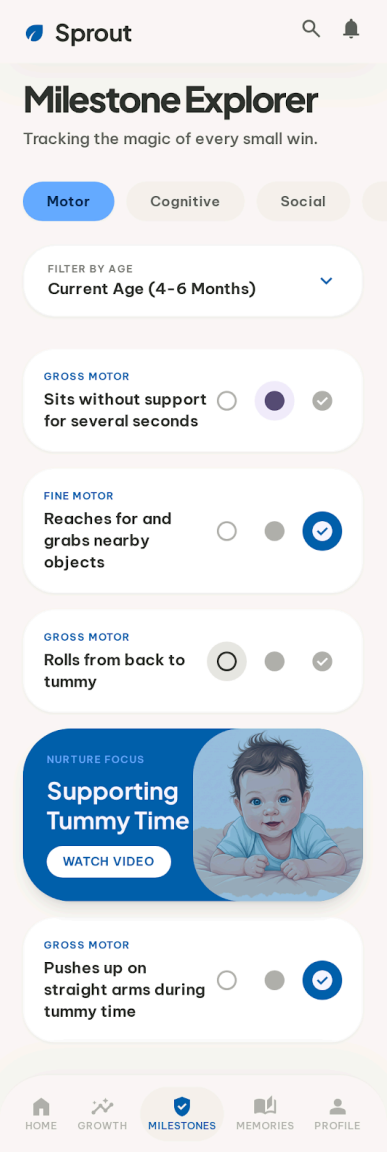
Screen 4: Growth Tracker (The Core WHO/CDC Engine)

- **Purpose:** Map physical development against clinical percentiles.
- **Developer Notes - The Action of "Logging Growth":** When a user "logs growth," they submit a payload containing the Date, Weight, Height, and Head Circumference. The backend must save this as a new row in the DB, run the numbers through the WHO/CDC LMS algorithm (using static CSV lookup tables based on the baby's exact age and sex), calculate the percentile, and instantly plot a new dot on the visual graph.
- **Features:**
    - **Visual Curve:** A line graph charting the baby's logged data points over the standard percentile bands (3rd, 15th, 50th, 85th, 97th percentiles).
    - **Data Entry Modal:** Input fields for Weight (kg/lbs), Height (cm/in), and Head Circumference (cm/in).
    - **Smart Flagging (Passive):** A simple logic check. If the newly calculated percentile drops two major bands lower than the previous entry, display a passive UI alert suggesting a doctor consult.



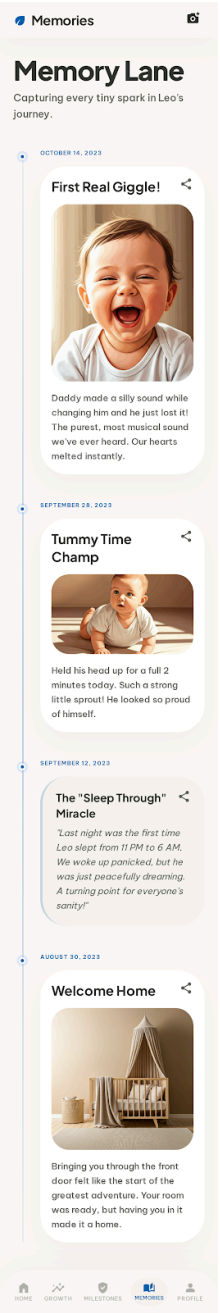
Screen 5: Milestone Explorer

- **Purpose:** A categorized, age-filtered checklist of developmental expectations to replace Google searching.
- **Developer Notes - The Action of "Logging a Milestone":** The app relies on a static JSON database of CDC milestones (e.g., ID: 104, Skill: "Sits without support", Category: Motor, Range: 6-9 months). "Logging" means the user updates the status of a specific skill ID. This pushes an update to the user's DB profile, changing the state from `null` to `achieved` and appending a timestamp.
- **Features:**
    - **Category Tabs:** Motor, Cognitive, Social, Language.
    - **Age Filters:** Defaults to showing milestones matching the baby's current calculated age block ("Current Age", "Past", "Upcoming").
    - **Milestone Cards:** Displays the description and the typical age range.
    - **Status Toggles:** Users select ✅ Achieved, 🟡 In progress, or ⏳ Not yet. Selecting "Achieved" triggers an optional prompt to attach a photo or note.



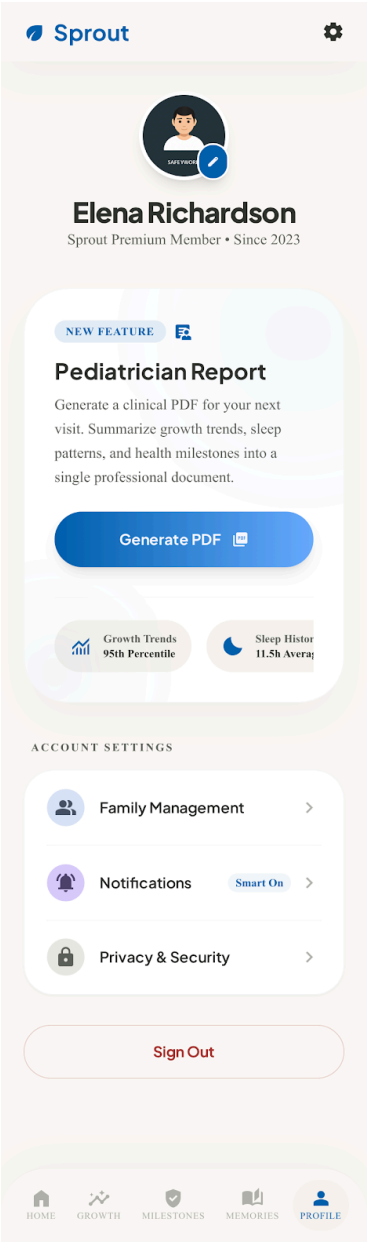
Screen 6: Memory Lane (The Emotional Hook)

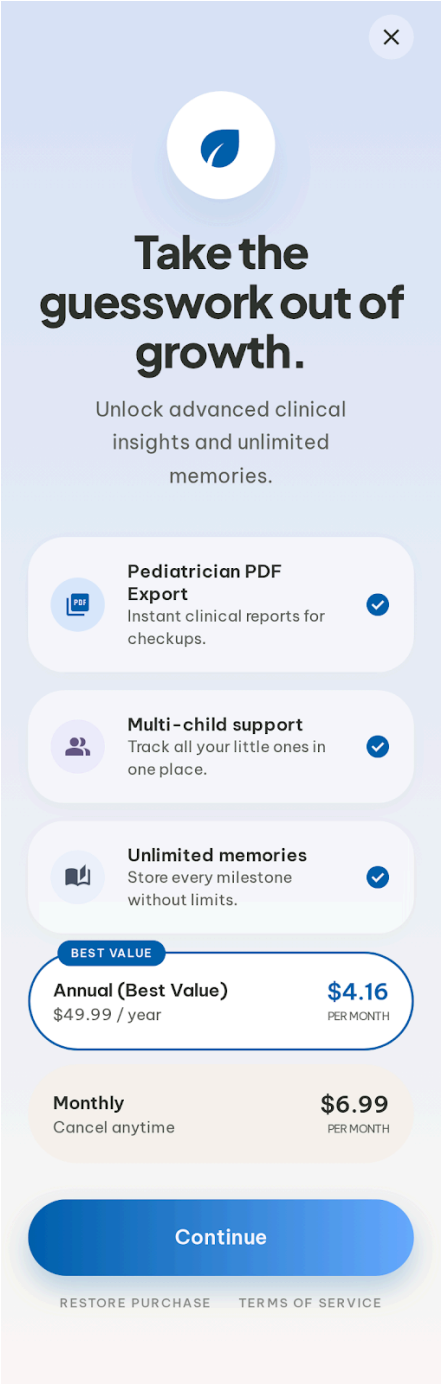
- **Purpose:** A chronological timeline creating a digital baby book of achievements.
- **Developer Notes:** A standard feed architecture. Query the user's milestone database for all entries where `status = achieved`, sorted by `date_achieved` in descending order.
- **Features:**
  - **Timeline View:** A vertical, scrolling feed of completed milestones.
  - **Memory Cards:** Displays the date, the specific milestone (e.g., "First Steps"), and any attached photo or parent text note.
  - **Share Button:** Native iOS/Android share sheet integration to easily send a specific memory card to family via text or social media.



Screen 7: Profile & Pediatrician Export

- **Purpose:** Account management and delivering the high-value, defensible PDF report feature.
- **Developer Notes:** The developer will need a robust PDF generation library to take the graph view from Screen 4 and the achieved/delayed list from Screen 5 and render them into a clean, printable document layout.
- **Features:**
    - **Family Management:** Add a second child (Premium) or manage partner access (via deep link or email invite).
    - **Pediatrician Report (The Hook):** A button to "Generate PDF Report." This compiles recent growth data, current percentiles, and milestone status into a standard 1-page format for offline viewing or printing.
    - **Settings:** Standard notification and privacy controls.



Screen 8: Paywall (Sprout Premium)

- **Purpose:** Upselling users to the premium tier to unlock full capabilities.
- **Developer Notes:** Standard implementation using tools like RevenueCat to manage entitlement states.
- **Features:**
  - **Premium Locks:** Checks user entitlement before allowing access to the Pediatrician PDF Export, multi-child support, and adding unlimited photos to Memory Lane.
  - **Pricing Cards:** Interactive cards for Annual and Monthly subscription tiers.